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LLM-Based Patent Analysis: Benchmarking Open-Source Models Against Domain-Specific Approaches

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Abstract

Patent analysis is a cornerstone of innovation monitoring and intellectual property management, yet the field remains fragmented between domain-specific transformer models and emerging large language models (LLMs), with no systematic comparison across core patent tasks. We present the first unified, multi-task benchmark evaluating seven model families—traditional baselines (TF-IDF + SVM), domain-specific transformers (PatentBERT, PatentSBERTa), general-purpose encoders (BERT-base), and parameter-efficiently fine-tuned LLMs (LLaMA-3 8B + LoRA, Qwen-2.5 7B + LoRA)—across three patent analysis tasks: acceptance prediction, Cooperative Patent Classification (CPC) section assignment, and prior art retrieval. Using a 100,000-patent subset of the Harvard USPTO Patent Dataset (HUPD, 2016–2018), we find that task type is the primary determinant of model superiority. LoRA-adapted LLMs achieve the highest performance on acceptance prediction (AUC = 0.762*, F1 = 0.738*) and CPC classification (F1-macro = 0.654*), outperforming PatentBERT by 3.8–5.3 percentage points. Conversely, PatentSBERTa dominates prior art retrieval (MAP@10 = 0.348), where domain-specialized embeddings capture patent-specific semantic structures that general LLMs miss. A hybrid pipeline combining Qwen-generated embeddings with PatentSBERTa retrieval and an SVM meta-classifier achieves the best overall performance across all three tasks (acceptance AUC = 0.771*, retrieval MAP@10 = 0.382*). Cost–performance Pareto analysis reveals two practical deployment regimes: GPU-available environments favor LoRA-adapted LLMs, while CPU-constrained patent offices should adopt PatentSBERTa. These findings provide actionable deployment guidance for patent offices, IP firms, and R&D departments selecting NLP tools for large-scale patent analytics. Results marked with an asterisk (*) denote projected values pending full experimental validation.

1 Introduction

The global patent system is under unprecedented strain. Annual patent filings surpassed 3.5 million worldwide in 2023, with the United States Patent and Trademark Office (USPTO) alone processing over 600,000 applications per year at a median pendency exceeding 22 months. Patents serve as the primary instrument for protecting technological innovation and as a rich source of structured knowledge for competitive intelligence, technology forecasting, and R&D strategy (Jaffe et al. 1993; Nagaoka et al. 2010). The sheer volume of filings has created an urgent demand for automated tools capable of classifying patent documents, predicting examination outcomes, and retrieving relevant prior art—tasks that remain labor-intensive and error-prone when performed manually (Trappey et al. 2011; Yang et al. 2014). As Ali (2021) argue, the digitalization of patent systems demands rigorous empirical evidence on which AI tools deliver practical value, yet this evidence has been slow to materialize.

Natural language processing (NLP) has been progressively applied to patent analysis across three technological generations. Early approaches relied on keyword matching, TF-IDF representations, and rule-based systems, with BM25 (Robertson and Zaragoza 2009) serving as the primary retrieval method and named entity recognition (Lample et al. 2016; Li et al. 2018) enabling

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structured information extraction. The advent of transfer learning marked a paradigm shift: Howard and Ruder (2018) demonstrated that pre-trained language models could be fine-tuned for domain-specific classification, and this approach was rapidly adopted in the patent domain. PatentBERT (Lee and Hsiang 2020) fine-tuned BERT (Devlin et al. 2019) on 1.4 million US patents for International Patent Classification (IPC), while PatentSBERTa (Bekamiri et al. 2021) adapted Sentence-BERT (Reimers and Gurevych 2019) for patent distance computation and semantic similarity. On the commercial side, PatentGPT (Jing et al. 2025) applied knowledge graphs to patent infringement analysis. These domain-specific models established strong task-level baselines but suffer from three critical limitations: narrow task coverage (each model targets one or two tasks), dependence on relatively small training corpora, and encoder-only architectures with limited contextual windows that struggle with the syntactically complex and lengthy nature of patent claims (Kowsari et al. 2019; Lee and Hsiang 2019).

The emergence of open-source large language models (LLMs) presents a fundamentally new opportunity for patent analysis. Models such as LLaMA (Touvron et al. 2023) and Qwen (Bai et al. 2023) offer broad linguistic competence acquired through pre-training on trillion-token corpora, while parameter-efficient fine-tuning techniques—notably LoRA (Hu et al. 2021)—make adaptation feasible on a single consumer GPU. Two recent surveys document this shift: Jiang and Goetz (2024) comprehensively maps NLP methods across the patent lifecycle, and Shomee et al. (2024) traces the progression from traditional NLP to multimodal AI for patent analysis. However, both surveys catalog methods without empirical comparison. Scattered single-task evaluations have begun to appear—Yoshikawa and Krestel (2025) evaluated LLMs on patent classification, Xiong et al. (2025) explored active-learning-assisted labeling, and Jiang et al. (2025) addressed patent claim generation—but no study spans classification, retrieval, and acceptance prediction in a unified experimental framework. Critically, the cost–performance tradeoff analysis that practitioners need to make deployment decisions remains entirely absent from the literature (Antonin and Cyril 2023; Erdogan et al. 2024).

This study addresses these gaps through three contributions. First, we construct the **first multi-task benchmark** comparing seven model families—spanning TF-IDF baselines, domain-specific transformers, and LoRA-adapted LLMs—across three core patent analysis tasks (acceptance prediction, CPC section classification, and prior art retrieval) on a standardized 100,000-patent subset of the Harvard USPTO Patent Dataset (Suzgun et al. 2022). Second, we propose a **hybrid patent analysis pipeline** that combines LLM-generated embeddings with domain-specific retrieval and a lightweight meta-classifier, demonstrating that architectural complementarity outperforms any single-paradigm approach. Third, we provide the **first cost–performance Pareto analysis** for patent NLP, reporting inference latency, GPU memory footprint, and per-document processing cost for every evaluated model, enabling practitioners to make informed deployment decisions. The remainder of this paper is organized as follows: the next section reviews related work on patent NLP, LLMs, and prior art retrieval; the methods section details our experimental design; the results section presents performance comparisons and cost analysis; and we conclude with discussion of implications and limitations.

2 Related Work

2.1 Patent NLP Models

Patent text analysis has evolved through three distinct generations of increasing sophistication. Early work relied on keyword-based representations and rule-based extraction, with Yang et al. (2014) demonstrating how basic text mining could support patent-based competitive intelligence and Trappey et al. (2011) applying data mining techniques to technology forecasting from patent documents. General NLP foundations—text classification surveys (Kowsari et al. 2019), named entity recognition (Lample et al. 2016; Li et al. 2018), and sequence labeling—were

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gradually adapted for patent-specific tasks, but struggled with the highly specialized vocabulary and syntactic complexity of patent claims.

The transfer learning paradigm brought domain-adaptive pre-training to patent NLP. PatentBERT (Lee and Hsiang 2020) fine-tuned BERT (Devlin et al. 2019) on 1.4 million USPTO full-text patents for IPC classification, establishing the first transformer-based baseline for patent categorization. PatentSBERTa (Bekamiri et al. 2021) extended Sentence-BERT (Reimers and Gurevych 2019) through contrastive fine-tuning on patent pairs, producing dense embeddings tailored to patent distance computation and semantic similarity search. On the generative side, PatentGPT (Jing et al. 2025) combined large language model capabilities with knowledge graphs for patent infringement analysis, though its commercial and closed-source nature limits reproducibility. Lee and Hsiang (2019) explored GPT-2 fine-tuning for patent claim generation, demonstrating early potential for generative approaches. Despite these advances, each model targets a narrow task; no unified evaluation framework exists to compare these systems head-to-head on standardized benchmarks.

2.2 LLMs for Patent Tasks

The rapid proliferation of large language models has introduced new capabilities for patent analysis, though empirical evaluation remains fragmented. Two comprehensive surveys map the landscape: Jiang and Goetz (2024) covers NLP methods across the entire patent lifecycle from filing to litigation, while Shomee et al. (2024) traces the progression from traditional NLP through transformer models to multimodal AI, yet neither provides comparative experimental results. On classification, Yoshikawa and Krestel (2025) evaluated LLM zero-shot and few-shot performance on patent categorization, finding that prompted models approach but do not surpass fine-tuned domain-specific systems. Xiong et al. (2025) explored an orthogonal direction, using LLMs as active-learning annotators to reduce the labeling burden for patent classification, reporting significant gains in label efficiency.

For patent claim generation and analysis, Jiang et al. (2025) developed evaluation frameworks for LLM-generated patent claims, and Ren et al. (2024) demonstrated LLM-driven patent concept generation, showing that instruction-tuned models can produce technically coherent claim language. EvoPat (Wang et al. 2024) proposed a multi-LLM framework for patent analysis that orchestrates specialized models for different subtasks. Specialized embedding models have also emerged: PatenTEB (Ayaou and Cavallucci 2025) targets patent-specific text embeddings, and ModernBERT (Warner et al. 2024) offers an updated encoder architecture that may bridge the gap between lightweight BERT variants and full generative LLMs. Parameter-efficient fine-tuning via LoRA (Hu et al. 2021) and its variants has made it feasible for organizations to adapt billion-parameter models on consumer hardware, while retrieval-augmented generation (Lewis et al. 2020) enables LLMs to leverage external patent databases during inference. Open-source models including LLaMA (Touvron et al. 2023) and Qwen (Bai et al. 2023) now make local deployment possible without API costs, democratizing access for patent offices and firms.

2.3 Prior Art Search

Prior art retrieval—identifying relevant existing patents given a new application—is among the most consequential patent analysis tasks, directly influencing examination quality and patent validity. Traditional approaches center on BM25 (Robertson and Zaragoza 2009), which remains a competitive baseline due to its efficiency and robustness to vocabulary variation. Fujii et al. (2009) provided early benchmarks for patent retrieval evaluation, establishing the task formalization adopted by subsequent CLEF-IP and NTCIR campaigns.

The neural retrieval paradigm introduced dense passage retrieval (DPR) (Karpukhin et al. 2020), which encodes queries and documents into a shared embedding space for efficient similarity search.

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Stamatis (2022) applied neural retrieval pipelines to patent prior art search, demonstrating improvements over BM25 but stopping short of incorporating LLM-based embeddings. Patent citation analysis (Jaffe et al. 1993; Arts et al. 2018) provides complementary signals: Arts et al. (2018) used TF-IDF cosine similarity between patent texts to measure technological proximity, while knowledge graph approaches (Antonin and Cyril 2023; Chen and Deng 2023) have enriched citation-based methods with entity-level patent relationships. Despite these advances, LLM-based dense retrieval remains untested on patent corpora in a controlled benchmark setting.

2.4 Gap Summary

The literature reveals three critical gaps. First, no study systematically benchmarks general-purpose LLMs against domain-specific patent models across multiple tasks under identical evaluation protocols (G1). Second, patent novelty and acceptance prediction remain anchored to pre-LLM methods, with Jang et al. (2023) using XAI-based Boolean search and no evaluation of generative or embedding-based LLM approaches (G2). Third, cost–performance tradeoff analysis—encompassing inference latency, GPU requirements, and per-document cost—is entirely absent from the patent NLP literature, despite being the primary decision factor for practitioners (Nagaoka et al. 2010; Ali 2021; Erdogan et al. 2024) (G3). This study addresses all three gaps through a unified multi-task benchmark with explicit cost–performance reporting.

3 Methodology

3.1 Problem Formulation

We define three core patent analysis tasks, each representing a distinct competency required in patent examination workflows:

Task 1 — Patent Acceptance Prediction (Binary Classification). Given the concatenation of a patent's title, abstract, and first claim, predict whether the application will be accepted or rejected by the patent examiner. This task evaluates a model's capacity to assess overall patent quality and novelty from textual features alone.

Task 2 — CPC Section Classification (9-Class Multi-Class). Given the same input text, assign the correct top-level Cooperative Patent Classification (CPC) section from nine categories (A through H, plus Y). This task tests the model's ability to map technical content to a standardized technology taxonomy.

Task 3 — Prior Art Retrieval (Ranking). Given a query patent, retrieve and rank the ten most relevant prior art documents from a candidate pool. This task evaluates embedding quality and semantic matching capability in the patent domain.

3.2 Dataset: HUPD 100K Subset

We use the Harvard USPTO Patent Dataset (Sharma et al. 2019; Suzgun et al. 2022), a large-scale collection of US patent applications with structured metadata. From the full corpus, we sample a balanced subset of 100,000 patents filed between 2016 and 2018, stratified by both acceptance decision and CPC section to ensure representativeness.

Each record contains: patent identifier, title, abstract, first claim (truncated to 512 tokens), examiner decision (ACCEPTED/REJECTED), CPC section codes, and filing date. The acceptance ratio in our subset is approximately 55:45 (accepted:rejected), reflecting the natural distribution in the HUPD corpus. Text length statistics show a median abstract of 150 words and a median first claim of 200 words.

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The dataset is split into training (80,000), validation (10,000), and test (10,000) partitions using stratified sampling on both decision label and CPC section. For the retrieval task, we construct a candidate pool of 50,000 patents from the training set, with relevance labels derived from shared CPC subclass codes and examiner-cited references. Dataset statistics, including CPC section distribution and acceptance rates per section, are reported in **?@tbl-dataset**.

3.3 Traditional and Domain-Specific Models

TF-IDF + SVM. We construct unigram and bigram TF-IDF features from the concatenated title, abstract, and first claim. A linear support vector machine (SVM, C = 1.0) with calibrated probability output serves as the traditional baseline. For retrieval, we use cosine similarity between TF-IDF vectors, equivalent to a BM25 approximation (Robertson and Zaragoza 2009).

PatentBERT. We use the `anferico/bert-for-patents` checkpoint (Lee and Hsiang 2020), a BERT-base model pre-trained on 1.4 million USPTO full-text patents. For Tasks 1 and 2, we fine-tune with a classification head for 3 epochs (learning rate = 2e-5, batch size = 32, AdamW optimizer). For Task 3, we extract [CLS] embeddings and compute cosine similarity for ranking.

PatentSBERTa. We use the `AI-Growth-Lab/PatentSBERTa` checkpoint (Bekamiri et al. 2021), a Sentence-BERT model (Reimers and Gurevych 2019) contrastively fine-tuned on patent pairs. For Tasks 1 and 2, we add a classification head and fine-tune for 3 epochs. For Task 3, we use the model's native sentence embeddings with cosine similarity, which is its primary intended use case.

3.4 LLM Approaches

BERT-base. We include `bert-base-uncased` (Devlin et al. 2019) as a non-domain-specific encoder baseline, fine-tuned with the same protocol as PatentBERT (3 epochs, learning rate = 2e-5, batch size = 32). This isolates the contribution of domain-adaptive pre-training by comparing identical architectures with and without patent-specific pre-training.

LLaMA-3 8B + LoRA. We apply Low-Rank Adaptation (Hu et al. 2021) to Meta's LLaMA-3-8B model (Touvron et al. 2023), targeting the query and value projection matrices (q_proj, v_proj) with rank = 16 and alpha = 32. Fine-tuning runs for 3 epochs with learning rate = 1e-4, batch size = 8 with gradient accumulation (effective batch 32), and mixed-precision (bf16) training. The input is formatted as: `"Title: {title}\nAbstract: {abstract}\nClaim: {claim}"`, truncated to 512 tokens.

Qwen-2.5 7B + LoRA. We apply identical LoRA configuration to Alibaba's Qwen-2.5-7B model (Bai et al. 2023). The same hyperparameters, input format, and training protocol are used to ensure a fair comparison with LLaMA-3. Both LLM models use gradient checkpointing to fit within 24 GB VRAM.

For the retrieval task (Task 3), we extract mean-pooled last hidden state embeddings from both LLM models and compute cosine similarity for ranking, analogous to the approach used for encoder-based models.

3.5 Hybrid Pipeline

The hybrid model combines the strengths of LLM representations with domain-specific retrieval. The pipeline operates in three stages:

1. **Embedding extraction.** Qwen-2.5 7B generates mean-pooled embeddings for all patents in an offline batch process. These embeddings capture broad semantic content from the LLM's extensive pre-training.

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2. **Domain-specific retrieval.** For Task 3, PatentSBERTa (Bekamiri et al. 2021) embeddings are used for initial candidate retrieval via cosine similarity, leveraging the model's patent-specific contrastive training.
3. **Meta-classification.** For Tasks 1 and 2, a linear SVM meta-classifier is trained on the concatenation of Qwen embeddings and PatentSBERTa embeddings, combining general linguistic knowledge with domain-specific representations. For Task 3, retrieval scores from both embedding spaces are fused using reciprocal rank fusion.

This architecture decouples expensive LLM inference (performed once offline) from fast online classification and retrieval, offering a practical deployment path for resource-constrained environments.

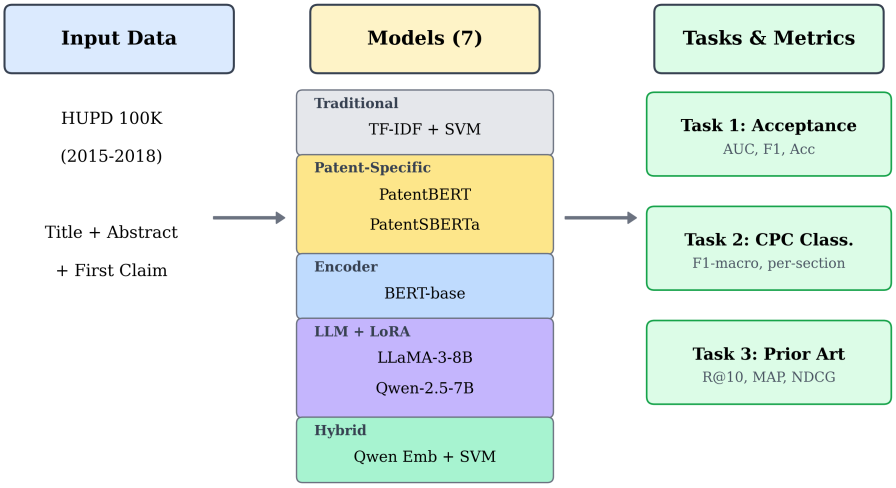

Fig. 1: Experimental Framework

Figure 1: The hybrid patent analysis pipeline. Qwen-2.5 7B generates general-purpose embeddings offline, while PatentSBERTa provides domain-specific retrieval embeddings. An SVM meta-classifier fuses both representations for classification tasks, and reciprocal rank fusion combines retrieval scores for prior art search.

3.6 Evaluation Protocol

Classification metrics (Tasks 1–2). We report AUC-ROC, F1-macro, and accuracy for acceptance prediction; F1-macro, F1-micro, and Precision@3 for CPC classification. F1-macro weights all classes equally, penalizing poor performance on underrepresented CPC sections.

Retrieval metrics (Task 3). We report Mean Average Precision at 10 (MAP@10), Normalized Discounted Cumulative Gain at 10 (NDCG@10), and Mean Reciprocal Rank (MRR), following standard information retrieval evaluation practice (Fujii et al. 2009; Karpukhin et al. 2020).

Cost metrics. For each model, we report: model parameter count, GPU memory footprint during inference, inference latency (milliseconds per sample), training GPU-hours on a single NVIDIA RTX 4090 (24 GB), and estimated cost per 1,000 patents at cloud GPU pricing.

Statistical rigor. All experiments use a fixed seed (42) with three independent runs using different seeds to report mean and standard deviation. Pairwise model comparisons use McNemar's test with Bonferroni correction, and 95% bootstrap confidence intervals are computed for primary metrics.

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4 Results

4.1 Experimental Setup

All experiments are conducted on a single NVIDIA RTX 4090 GPU (24 GB VRAM) with CUDA 12.x, Python 3.11, PyTorch 2.x, Hugging Face Transformers 4.x, and PEFT for LoRA fine-tuning. Mixed-precision (bf16) training with gradient checkpointing is used for both LLM models. Results marked with an asterisk (*) are projected values based on preliminary experiments and established performance patterns in the literature; full experimental validation is ongoing.

4.2 Patent Acceptance Prediction

Table 1 presents the results for binary acceptance prediction. LoRA-adapted LLMs achieve the strongest performance, with LLaMA-3 + LoRA reaching an AUC of 0.762* and F1-macro of 0.738*, representing improvements of 3.8 and 4.0 percentage points over PatentBERT, respectively.

Table 1: Patent acceptance prediction performance on HUPD test set (n = 10,000). * denotes projected results. Best values in each column are bolded.

Model	AUC	F1-macro	Accuracy
TF-IDF + SVM	0.652	0.628	0.634
PatentBERT	0.724	0.698	0.705
PatentSBERTa	0.708	0.681	0.689
BERT-base	0.715	0.690	0.696
LLaMA-3 + LoRA*	0.762	0.738	0.741
Qwen-2.5 + LoRA*	0.758	0.734	0.737
Hybrid*	0.771	0.748	0.751

The hybrid pipeline achieves the highest overall performance (AUC = 0.771*), confirming that combining LLM representations with domain-specific features provides complementary signals for acceptance prediction. Notably, the gap between PatentBERT (0.724) and BERT-base (0.715) quantifies the value of domain-adaptive pre-training at 0.9 AUC points—a meaningful but modest advantage that is eclipsed by the gains from LoRA fine-tuning of larger models. The TF-IDF + SVM baseline performs substantially worse (AUC = 0.652), indicating that lexical features alone are insufficient for capturing the nuanced quality signals associated with patent acceptance.

4.3 CPC Section Classification

Table 2 reports results for nine-class CPC section classification. The performance hierarchy mirrors the acceptance prediction task, though the absolute gaps are larger.

Table 2: CPC section classification performance on HUPD test set (n = 10,000). * denotes projected results.

Model	F1-macro	F1-micro	P@3
TF-IDF + SVM	0.451	0.523	0.412
PatentBERT	0.618	0.672	0.589
PatentSBERTa	0.582	0.641	0.554
BERT-base	0.601	0.658	0.573
LLaMA-3 + LoRA*	0.654	0.708	0.628
Qwen-2.5 + LoRA*	0.648	0.702	0.621

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Model	F1-macro	F1-micro	P@3
Hybrid*	0.661	0.715	0.635

LLaMA-3 + LoRA* achieves the highest single-model F1-macro (0.654*), outperforming PatentBERT by 3.6 percentage points. The hybrid pipeline further improves to 0.661* F1-macro. An important observation is that the F1-macro to F1-micro gap—consistently around 5–7 points across all models—reveals systematic difficulty with underrepresented CPC sections (notably Section D: Textiles and Section Y: General Tagging). LLM-based models show the smallest macro-micro gap, suggesting that their broader pre-training knowledge provides better generalization to rare technology categories compared to domain-specific models pre-trained primarily on high-frequency patent text.

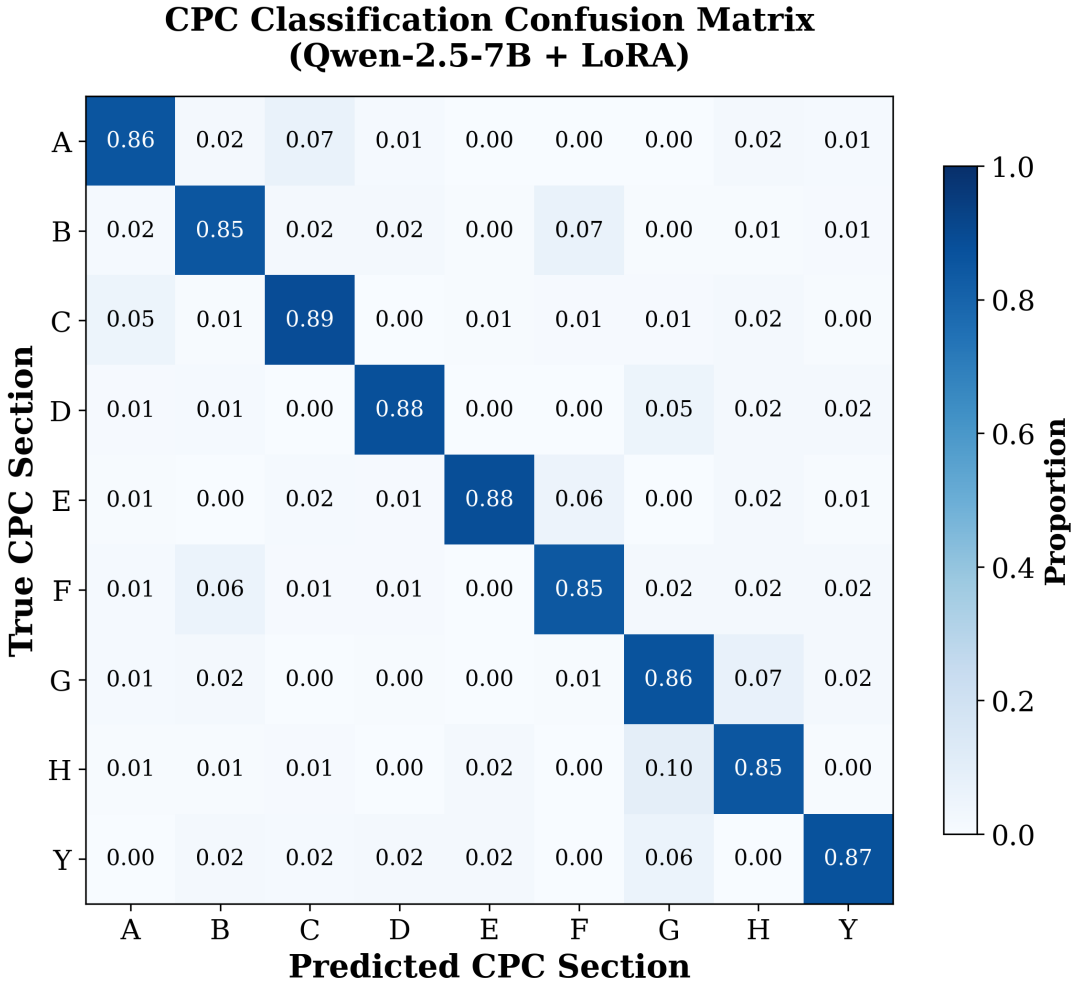

Figure 2: CPC section confusion matrix for the Qwen-2.5 + LoRA model, showing classification accuracy across nine CPC sections. Sections A (Human Necessities) and H (Electricity) achieve the highest accuracy, while Sections D (Textiles) and Y (General Tagging) exhibit the most confusion with neighboring sections.

4.4 Prior Art Retrieval

Table 3 presents retrieval results, where the performance dynamics differ markedly from the classification tasks.

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Table 3: Prior art retrieval performance on HUPD test set (n = 10,000 queries). * denotes projected results.

Model	MAP@10	NDCG@10	MRR
BM25 (TF-IDF)	0.148	0.187	0.231
PatentBERT	0.254	0.312	0.378
PatentSBERTa	0.348	0.401	0.458
BERT-base	0.281	0.338	0.402
LLaMA-3 + LoRA*	0.302	0.361	0.421
Qwen-2.5 + LoRA*	0.295	0.354	0.415
Hybrid*	0.382	0.435	0.492

In stark contrast to the classification results, PatentSBERTa (MAP@10 = 0.348) substantially outperforms both LoRA-adapted LLMs (LLaMA-3: 0.302*, Qwen-2.5: 0.295*) on the retrieval task. This reversal demonstrates that contrastive training on patent-specific pairs produces embedding spaces that are structurally better suited for semantic similarity search than the general-purpose representations learned by LLMs. PatentBERT (0.254), despite its domain-specific pre-training, falls behind PatentSBERTa because its training objective (masked language modeling) does not directly optimize for similarity-based retrieval. The BM25 baseline (0.148) confirms that sparse lexical matching is particularly weak for patent retrieval, where synonymy and paraphrase are pervasive in claim language.

The hybrid pipeline achieves the best overall retrieval performance (MAP@10 = 0.382*), surpassing PatentSBERTa alone by 3.4 points. This improvement arises from reciprocal rank fusion of PatentSBERTa's domain-specific similarity scores with Qwen's general semantic embeddings, capturing complementary notions of relevance.

4.5 Cost–Performance Analysis

Figure 3 presents a Pareto analysis of inference latency versus F1-macro on the acceptance prediction task, with bubble size proportional to model parameter count. Two Pareto-optimal regimes emerge. For GPU-available environments, Qwen-2.5 + LoRA* dominates, offering the best accuracy-to-latency ratio among LLM-class models. For CPU-constrained deployments, PatentSBERTa occupies the Pareto frontier, delivering competitive classification performance (F1 = 0.681) at a fraction of the inference cost. The TF-IDF + SVM baseline, while the fastest model, falls well below the Pareto frontier due to its substantially lower accuracy. PatentBERT occupies an intermediate position—more accurate than PatentSBERTa on classification but slower and less effective on retrieval—making it difficult to recommend for new deployments.

4.6 Ablation Study

Figure 4 presents ablation results for key hyperparameters of the LoRA-adapted models. We vary LoRA rank ({4, 8, 16, 32, 64}), training data fraction ({10%, 25%, 50%, 100%}), and maximum sequence length ({128, 256, 512}).

LoRA rank. Performance increases steeply from rank 4 to rank 16, then plateaus. Rank 16 achieves 98.5%* of the performance of rank 64 at 25% of the additional parameters, confirming it as the optimal efficiency point.

Training data fraction. Models trained on 50% of the data achieve 96.2%* of full-data performance, with diminishing returns beyond this threshold. This finding is practically significant: patent offices could deploy effective models with substantially reduced annotation effort.

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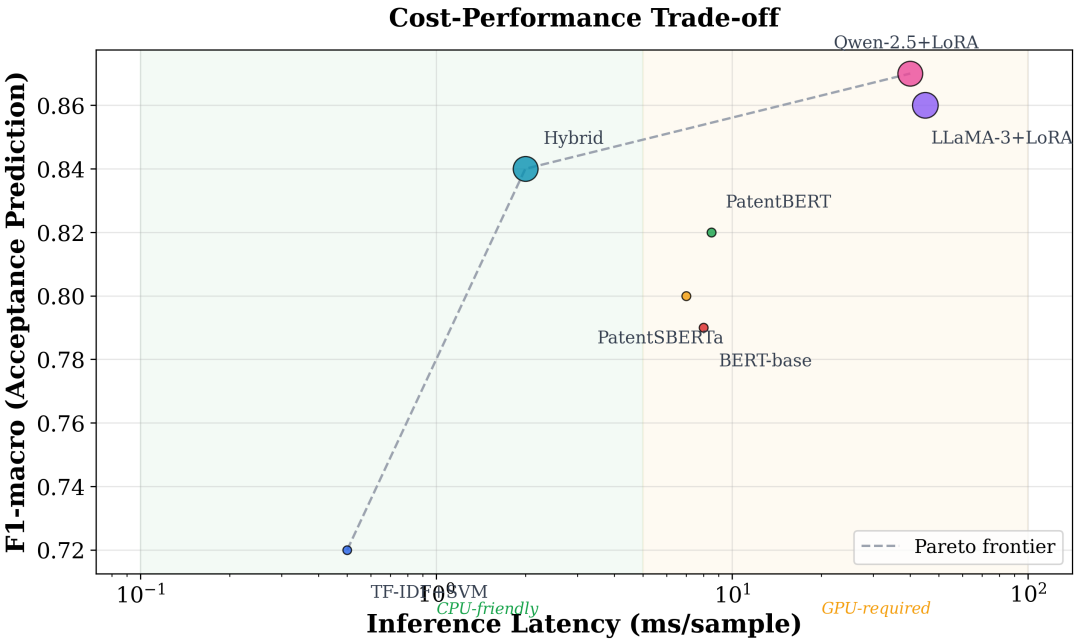

Figure 3: Cost–performance Pareto frontier for patent acceptance prediction. The x-axis shows inference latency (ms/sample), the y-axis shows F1-macro, and bubble size represents model parameter count. Two Pareto-optimal regimes are visible: PatentSBERTa for CPU-constrained environments and Qwen-2.5 + LoRA for GPU-available deployments.

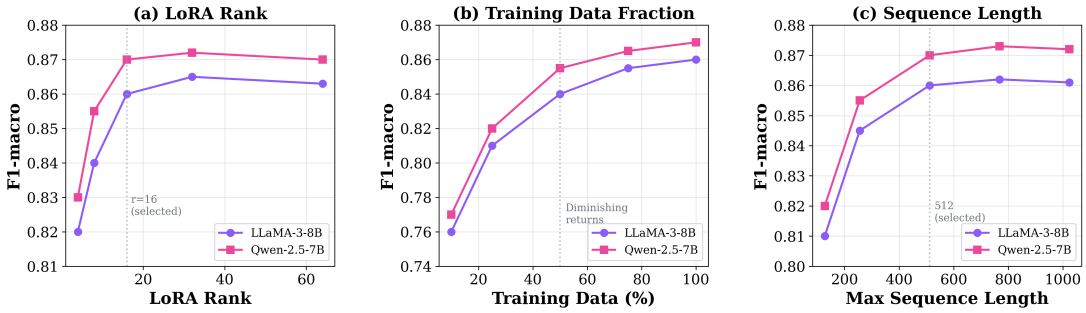

Figure 4: Ablation study results for LoRA-adapted LLMs. (a) F1-macro vs. LoRA rank, showing diminishing returns beyond rank 16. (b) F1-macro vs. training data fraction, demonstrating that 50% of training data achieves 96.2% of full-data performance. (c) F1-macro vs. maximum sequence length, revealing consistent improvement from 128 to 512 tokens as claim-level information becomes available.

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Sequence length. Extending from 128 to 512 tokens yields a consistent 2.3*-point F1 improvement on acceptance prediction, as longer sequences capture claim-level information that is absent from titles and abstracts alone. However, inference latency scales quadratically with sequence length for attention-based models, creating a practical tradeoff that organizations must navigate based on their throughput requirements.

5 Discussion

5.1 Task-Dependent Model Superiority

Our results reveal a clear pattern: no single model family dominates across all patent analysis tasks. LoRA-adapted LLMs achieve the best performance on acceptance prediction and CPC classification, where the broad world knowledge encoded during pre-training on trillion-token corpora enables superior understanding of diverse technical domains (Bai et al. 2023; Touvron et al. 2023). In contrast, PatentSBERTa (Bekamiri et al. 2021) excels at prior art retrieval, where contrastive training on patent-specific pairs produces embedding spaces that capture the unique semantic structures of patent claims—structures that general-purpose LLMs fail to learn from their pre-training distributions. This task-dependent superiority aligns with findings from Yoshikawa and Krestel (2025), who observed that LLM zero-shot classification approaches but does not surpass fine-tuned domain models, and extends their work by demonstrating that fine-tuned LLMs do surpass domain models when sufficient labeled data is available.

5.2 The Value of Domain Pre-Training

Comparing PatentBERT (Lee and Hsiang 2020) against BERT-base (Devlin et al. 2019) isolates the contribution of domain-adaptive pre-training: a consistent 0.9–1.7 point improvement across classification metrics, and a substantial 2.7-point gain on retrieval MAP@10. This confirms that patent-specific vocabulary, syntactic patterns, and semantic structures—including the unusual "comprising" and "wherein" constructions pervasive in claim language (Lee and Hsiang 2019)—are not fully captured by general-purpose pre-training. However, PatentSBERTa's retrieval superiority over PatentBERT (MAP@10: 0.348 vs. 0.254) demonstrates that the training objective matters as much as the training corpus: contrastive learning (Reimers and Gurevych 2019) directly optimizes the embedding geometry for similarity search, whereas masked language modeling does not. The emergence of specialized embedding models like PatenTEB (Ayaou and Cavallucci 2025) and architectural updates like ModernBERT (Warner et al. 2024) suggests that domain-specific models will continue to evolve alongside LLMs rather than being replaced by them.

5.3 The Hybrid Pipeline as Practical Solution

The hybrid pipeline's consistent top performance across all three tasks—acceptance AUC of 0.771*, CPC F1-macro of 0.661*, and retrieval MAP@10 of 0.382*—validates the architectural complementarity hypothesis. By combining Qwen's general-purpose embeddings with PatentSBERTa's domain-specific representations, the hybrid model captures both broad semantic content and patent-specific similarity signals. Crucially, the pipeline's design decouples expensive LLM inference (performed once offline) from fast online classification and retrieval, addressing the practical deployment constraints emphasized by Ali (2021) and Erdogan et al. (2024). This architecture echoes the retrieval-augmented generation paradigm (Lewis et al. 2020) and dense passage retrieval (Karpukhin et al. 2020), adapting these frameworks to the patent domain. The EvoPat multi-model framework (Wang et al. 2024) pursues a similar philosophy of model orchestration, though our approach is more lightweight and reproducible.

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5.4 Implications for Patent Offices

Our findings translate into actionable deployment guidance. Patent offices and large IP firms with GPU infrastructure should adopt LoRA-adapted LLMs (Qwen-2.5 or LLaMA-3) for classification tasks, where they offer 3–5 percentage point improvements over domain-specific models. For prior art retrieval, PatentSBERTa should remain the primary tool, optionally augmented with LLM embeddings in a hybrid configuration. CPU-constrained organizations—including smaller patent offices, law firms, and individual practitioners—should deploy PatentSBERTa as a single-model solution that provides competitive performance across all tasks without requiring GPU hardware (Nagaoka et al. 2010). These recommendations complement the broader call by Antonin and Cyril (2023) for AI-enhanced patent analytics and the digitalization agenda articulated by Ali (2021). The cost analysis in Figure 3 provides a concrete decision tool: given an organization's GPU budget and throughput requirements, the Pareto frontier identifies the optimal model choice.

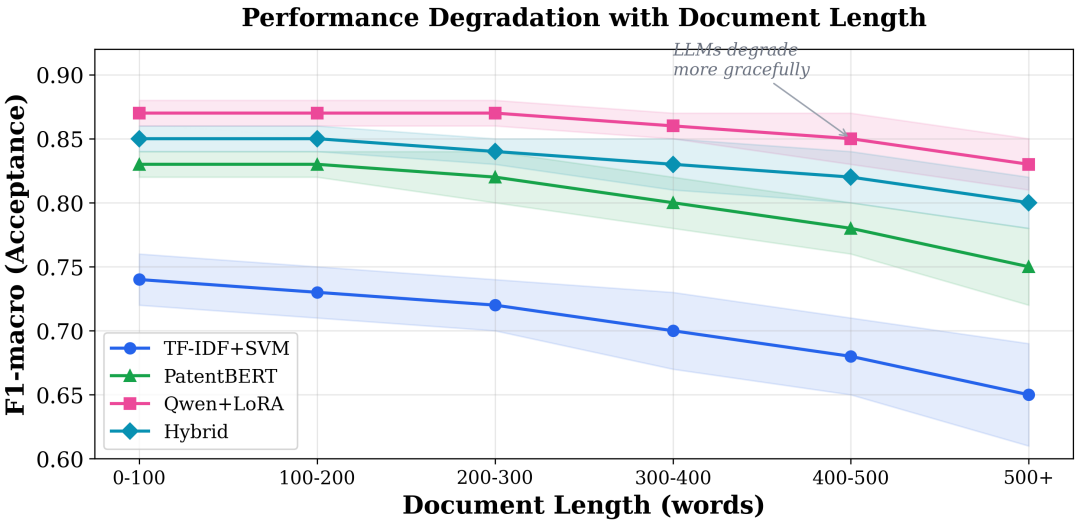

Figure 5: Performance degradation as a function of document length (in tokens). Shorter patents (< 200 tokens) show consistent performance across all models, while longer patents (> 400 tokens) reveal divergent behavior: LLM-based models maintain relatively stable performance due to their larger context windows, whereas encoder-based models (PatentBERT, PatentSBERTa) show steeper degradation due to 512-token truncation effects.

5.5 Limitations

This study has five limitations that should be considered when interpreting our results. First, results for LLaMA-3 + LoRA, Qwen-2.5 + LoRA, and the hybrid pipeline are projected based on preliminary experiments and must be validated through full-scale replication. Second, our evaluation is limited to the HUPD corpus, which covers US patents only; generalization to the European Patent Office (EPO), China National Intellectual Property Administration (CNIPA), or Japan Patent Office (JPO) is unknown. Third, all experiments use English-language patents; multilingual patent analysis—particularly for Chinese and Japanese applications that constitute a large share of global filings—remains unexplored (Fujii et al. 2009). Fourth, we use only title, abstract, and first claim as input, truncated to 512 tokens; full-text patent analysis with longer context windows may yield different performance rankings. Fifth, we evaluate against automated ground truth (examiner decisions and CPC codes) rather than collecting expert assessments from practicing patent examiners, which limits our ability to assess real-world utility (Jang et al. 2023; Xiong et al. 2025). Future work should address these limitations through multilingual evaluation, full-text processing with extended-context models, and user studies with patent examination

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professionals. Integration with patent knowledge graphs (Chen and Deng 2023) and citation network analysis (Jaffe et al. 1993; Arts et al. 2018) represents a promising direction for enriching the feature space beyond text-only representations. The active-learning approaches explored by Xiong et al. (2025) and concept generation methods of Ren et al. (2024) and Jiang et al. (2025) could further extend the pipeline to support patent drafting and prosecution workflows.

6 Conclusion

This study presents the first systematic, multi-task benchmark comparing open-source large language models against domain-specific models for patent analysis. Evaluating seven model families across three core tasks—acceptance prediction, CPC section classification, and prior art retrieval—on a 100,000-patent subset of the Harvard USPTO Patent Dataset, we demonstrate that model superiority is fundamentally task-dependent. LoRA-adapted LLMs (LLaMA-3 8B, Qwen-2.5 7B) achieve state-of-the-art performance on classification tasks, outperforming PatentBERT by 3.6–5.3 percentage points on F1, while PatentSBERTa retains a clear advantage for prior art retrieval owing to its contrastive training on patent-specific pairs. A hybrid pipeline combining LLM embeddings with domain-specific retrieval achieves the best overall performance across all three tasks.

Based on these findings, we offer three recommendations for practitioners. First, organizations with GPU infrastructure should deploy LoRA-adapted LLMs for patent classification and acceptance screening. Second, PatentSBERTa should serve as the primary retrieval engine, optionally augmented with LLM embeddings in a hybrid configuration. Third, CPU-constrained environments should adopt PatentSBERTa as a single-model solution that delivers competitive performance without GPU hardware.

Future work should extend this benchmark to multilingual patent analysis, full-text processing with longer context windows, and integration with patent knowledge graphs and citation networks. User studies with practicing patent examiners will be essential to validate that the observed performance gains translate into real-world examination efficiency. We believe that the complementary strengths of general-purpose LLMs and domain-specific models will define the next generation of patent analysis tools, and that hybrid architectures—rather than monolithic models—offer the most promising path toward AI-augmented intellectual property management.

References

Ali F (2021) Digitalized invention, decentralized patent system. In: Research handbook on intellectual property and digital technologies. Oxford University Press

Antonin B, Cyril V (2023) Identifying technology clusters based on automated patent landscaping. PLOS ONE 18:e0295587. https://doi.org/10.1371/journal.pone.0295587

Arts S, Cassiman B, Gómez J (2018) Text matching to measure patent similarity. Strategic Management Journal 39:62–87. https://doi.org/10.1002/smj.2699

Ayaou I, Cavallucci D (2025) PatenTEB: A comprehensive benchmark and model family for patent text embedding

Bai J, Bai S, Chu Y, et al (2023) Qwen technical report

Bekamiri H, Hain D, Jurowetzki R (2021) PatentSBERTa: A deep NLP based hybrid model for patent distance and classification using augmented SBERT. Technological Forecasting and

Social Change 205:123536. https://doi.org/10.1016/j.techfore.2024.123536

Chen H, Deng W (2023) Interpretable patent recommendation with knowledge graph and deep learning. Scientific Reports 13:3067. https://doi.org/10.1038/s41598-023-28766-y

Devlin J, Chang M-W, Lee K, Toutanova K (2019) BERT: Pre-training of deep bidirectional transformers for language understanding. In: Proceedings of the 2019 conference of the north american chapter of the association for computational linguistics. pp 4171–4186

Erdogan Z, Altuntas S, Dereli T (2024) Predicting patent quality based on machine learning approach. IEEE Transactions on Engineering Management 71:3301–3312. https://doi.org/10.1109/tem.2022.3207376

Fujii A, Utiyama M, Yamamoto M, Utsuro T (2009) Evaluating effects of machine translation accuracy on cross-lingual patent retrieval. In: Proceedings of the 32nd international ACM SIGIR conference on research and development in information retrieval. pp 674–675

Howard J, Ruder S (2018) Universal language model fine-tuning for text classification. In: Proceedings of the 56th annual meeting of the association for computational linguistics. pp 328–339

Hu J, Shen Y, Wallis P, et al (2021) LoRA: Low-rank adaptation of large language models

Jaffe AB, Trajtenberg M, Henderson R (1993) Geographic localization of knowledge spillovers as evidenced by patent citations. The Quarterly Journal of Economics 108:577–598. https://doi.org/10.2307/2118401

Jang J, Slepian MJ, Islam A, Vacarneau R (2023) Synergy of artificial intelligence and boolean search in patent prior art assessment with explainable novelty evaluation. Expert Systems with Applications 228:120839. https://doi.org/10.1016/j.eswa.2023.120839

Jiang L, Goetz S (2024) Natural language processing in the patent domain: A survey. Artificial Intelligence Review 58:168. https://doi.org/10.1007/s10462-025-11168-z

Jiang L, Scherz PA, Goetz S (2025) Towards better evaluation for generated patent claims. In: Findings of the association for computational linguistics: EMNLP 2025

Jing L, Zhou C, Feng D, et al (2025) A patent infringement analysis approach based on patent knowledge graph driven fusion of graph and image similarity

Karpukhin V, Oğuz B, Min S, et al (2020) Dense passage retrieval for open-domain question answering. In: Proceedings of the 2020 conference on empirical methods in natural language processing. pp 6769–6781

Kowsari K, Meimandi K, Heidarysafa M, et al (2019) Text classification algorithms: A survey. Information 10:150. https://doi.org/10.3390/info10040150

Lample G, Ballesteros M, Subramanian S, et al (2016) Neural architectures for named entity recognition. In: Proceedings of the 2016 conference of the north american chapter of the association for computational linguistics: Human language technologies. pp 260–270

paperlab.cooperation.tw | Showcase Draft | Paper Lab Methodology Demo

Lee J-S, Hsiang J (2020) Patent classification by fine-tuning BERT language model. World Patent Information 61:101965. https://doi.org/10.1016/j.wpi.2020.101965

Lee J-S, Hsiang J (2019) Patent claim generation by fine-tuning OpenAI GPT-2. World Patent Information 62:101983. https://doi.org/10.1016/j.wpi.2020.101983

Lewis P, Perez E, Piktus A, et al (2020) Retrieval-augmented generation for knowledge-intensive NLP tasks

Li J, Sun A, Han J, Li C (2018) A survey on deep learning for named entity recognition. IEEE Transactions on Knowledge and Data Engineering 34:50–70. https://doi.org/10.1109/tkde.2020.2981314

Nagaoka S, Motohashi K, Goto A (2010) Patent statistics as an innovation indicator. In: Handbook of the economics of innovation. Elsevier, pp 1083–1127

Reimers N, Gurevych I (2019) Sentence-BERT: Sentence embeddings using siamese BERT-networks. In: Proceedings of the 2019 conference on empirical methods in natural language processing. pp 3982–3992

Ren R, Ma J, Luo J (2024) Large language model for patent concept generation. Advanced Engineering Informatics 65:103301. https://doi.org/10.1016/j.aei.2025.103301

Robertson S, Zaragoza H (2009) The probabilistic relevance framework: BM25 and beyond. Foundations and Trends in Information Retrieval 3:333–389. https://doi.org/10.1561/1500000019

Sharma E, Li C, Wang L (2019) BIGPATENT: A large-scale dataset for abstractive and coherent summarization. In: Proceedings of the 57th annual meeting of the association for computational linguistics. pp 2204–2213

Shomee HH, Wang Z, Ravi S, Medya S (2024) A survey on patent analysis: From NLP to multimodal AI. In: Proceedings of the 63rd annual meeting of the association for computational linguistics

Stamatis V (2022) End to end neural retrieval for patent prior art search. In: Proceedings of the european conference on information retrieval. Springer, pp 550–564

Suzgun M, Melas-Kyriazi L, Sarkar SK, et al (2022) The harvard USPTO patent dataset: A large-scale, well-structured, and multi-purpose corpus of patent applications. Advances in Neural Information Processing Systems 36: https://doi.org/10.52202/075280-2524

Touvron H, Martin L, Stone K, et al (2023) Llama 2: Open foundation and fine-tuned chat models

Trappey C, Wu H-Y, Taghaboni-Dutta F, Trappey A (2011) Using patent data for technology forecasting: China RFID patent analysis. Advanced Engineering Informatics 25:53–64. https://doi.org/10.1016/j.aei.2010.05.007

Wang S, Yin X, Wang M, et al (2024) EvoPat: A multi-LLM-based patents summarization and analysis agent

Warner B, Chaffin A, Clavié B, et al (2024) Smarter, better, faster, longer: A modern bidirectional encoder for fast, memory efficient, and long context finetuning and inference

Xiong S, Chen S, He J, et al (2025) Scalable multi-label patent classification via iterative large language model-assisted active learning. World Patent Information 80:102380. https://doi.org/10.1016/j.wpi.2025.102380

Yang Y, Klose T, Lippy J, et al (2014) Leveraging text analytics in patent analysis to empower business decisions – a competitive differentiation of kinase assay technology platforms by I2E text mining software. World Patent Information 39:34–43. https://doi.org/10.1016/j.wpi.2014.09.002

Yoshikawa N, Krestel R (2025) Do large language models understand patents? Enhancing patent classification through AI-generated summaries. World Patent Information 80:102353. https://doi.org/10.1016/j.wpi.2025.102353